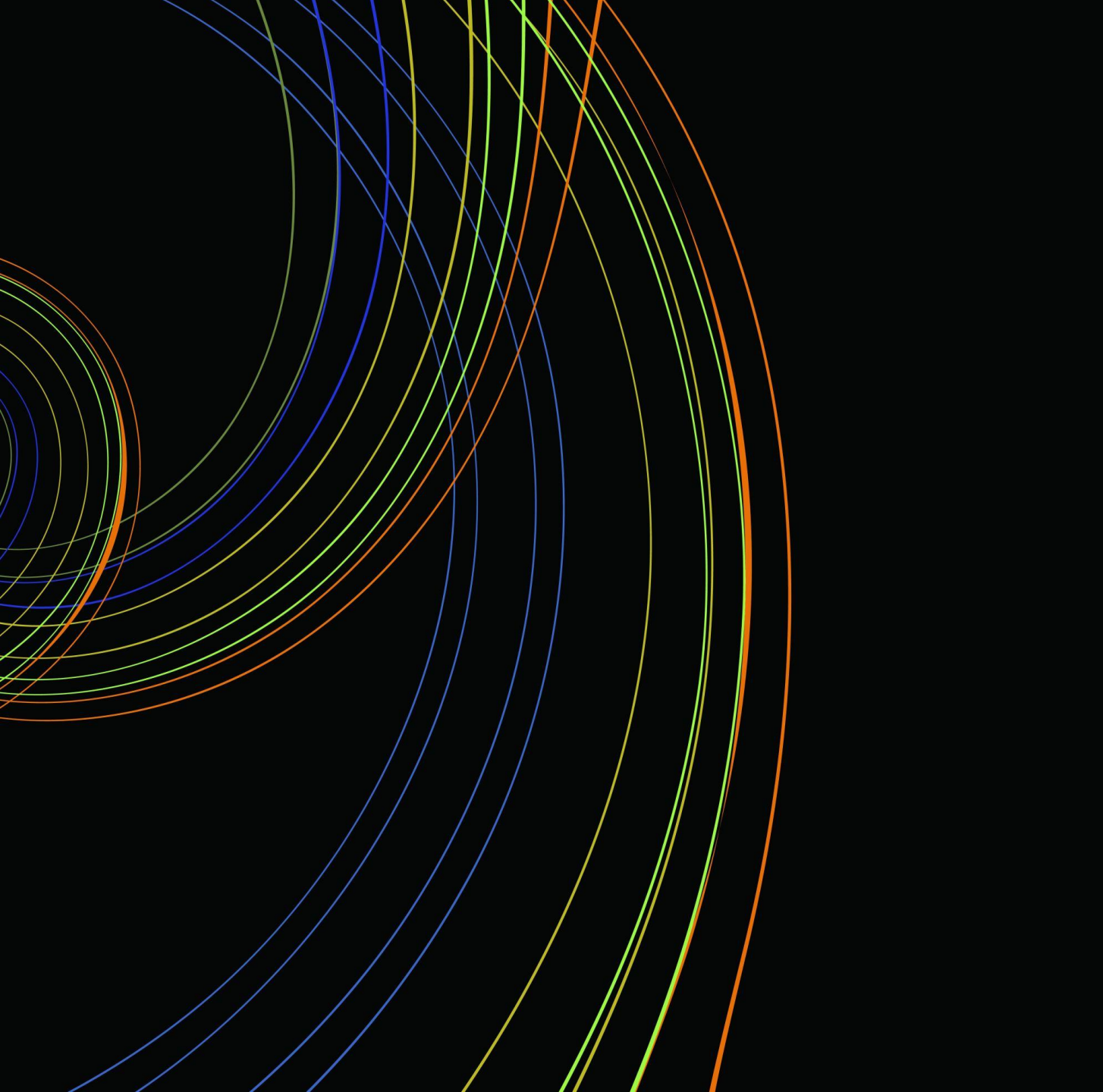




WEEK 4 ZKP



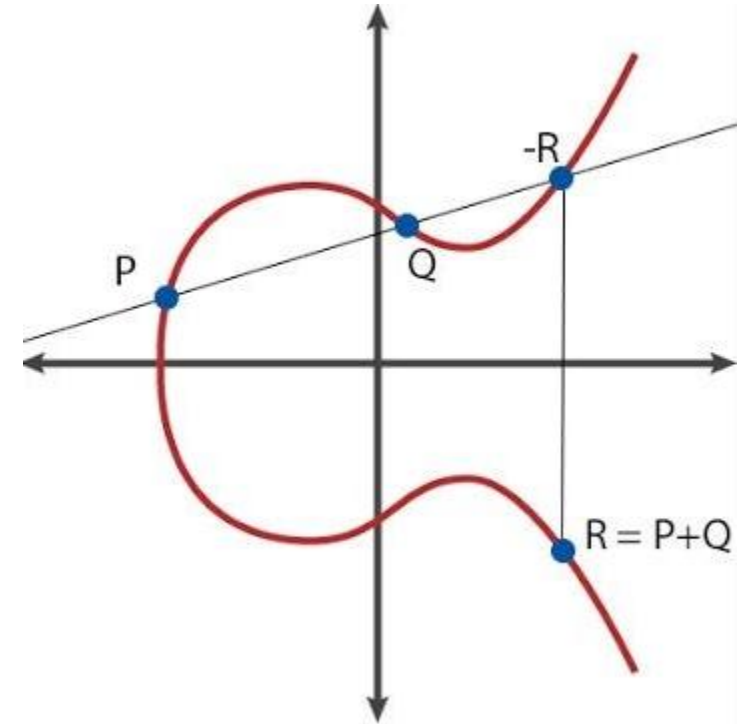
NITHESH

Elliptic Curve Group

- It is the group of points on a Set G that satisfies:

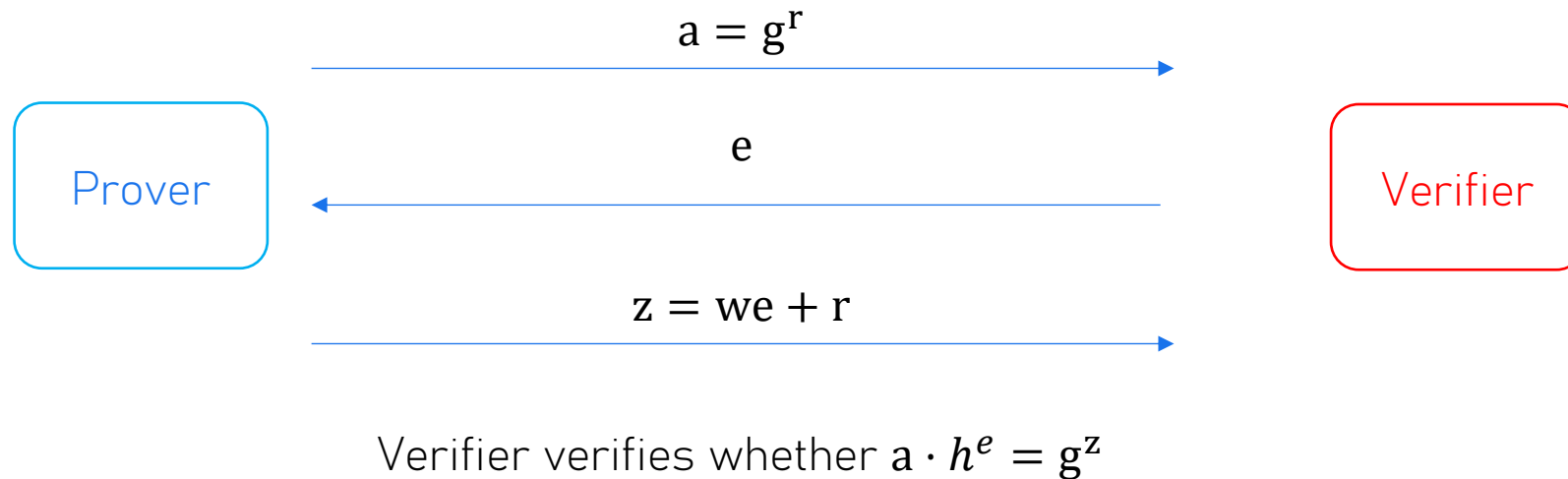
$$y^2 = x^3 + ax + b, \quad a, b \in \mathbb{R}$$

- Given two points P and Q , draw a line through them. The line intersects the curve at a third point R , which is reflected to define $P + Q$.
- Suppose $a, b \in G$ such that $b = xa$ where $x \in \mathbb{R}$. Given a and b , finding x is computationally expensive.



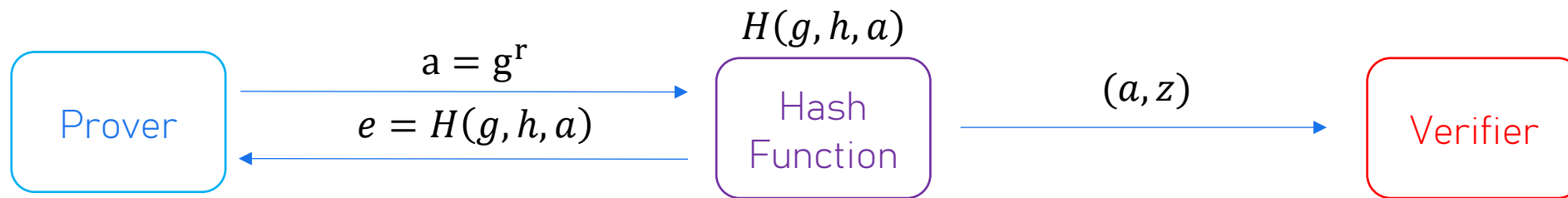
Schnorr's Σ -Protocol

- The Discrete Logarithm Problem:
Given group elements g and h , find w such that $h = g^w$
- The protocol is **3-message public-coin interactive proof** between a prover and a verifier. Where the prover proves knowledge of w such that $h = g^w$, without revealing w .



Apply Fiat-Shamir

- Makes the Σ -Protocol non-interactive
- Instead of the verifier sending a random challenge e , it is generated by a hash function $e = H(g, h, a)$.
- The proof can be sent, stored, and verified later by anyone



Verifier verifies whether $a \cdot h^e = g^z$

Homomorphic Commitment Scheme (HCS)

- The **commitment scheme** lets a prover commit to a value m without revealing it. $C = \text{Commit}(m, r)$ where m is the hidden message and r is randomness.
- Hiding property: where the verifier cannot learn the committed value m from C .
- Binding property: where after committing, the prover cannot change the commitment to a different value.
- Homomorphic: to get a commitment to the sum $m_1 + m_2$ without opening C_1 and C_2 we can just combine $C_1 \cdot C_2$

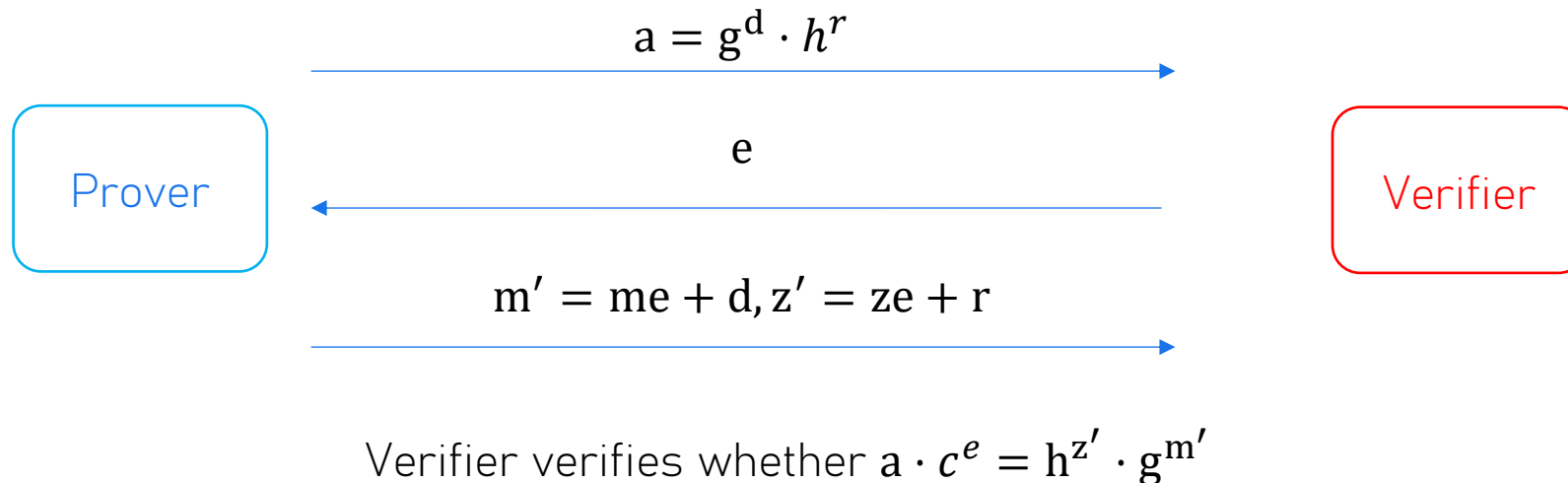
$$\text{Commit}(m_1, r_1) \cdot \text{Commit}(m_2, r_2) = \text{Commit}(m_1 + m_2, r_1 + r_2)$$

Pedersen Commitments

- Specific HCS where $C = \text{Commit}(m, z) = g^m \cdot h^z$
- perfectly hiding and computationally binding
- Homomorphic property:

$$c_1 = g^{m_1} \cdot h^{z_1}, c_2 = g^{m_2} \cdot h^{z_2}$$
$$c_1 c_2 = g^{m_1+m_2} \cdot h^{z_1+z_2} = \text{Commit}(m_1+m_2, z_1+z_2)$$

Proof of Knowledge of Opening :



Polynomial Commitment

The prover does not need to send the whole polynomial $q(X)$ to the verifier.

$$q(X) = u_0 + u_1X + u_2X^2 + \dots + u_{n-1}X^{n-1}$$
$$u = (u_0, u_1, \dots, u_{n-1}), \quad \text{For evaluation at point } z, y = (1, z, z^2, \dots, z^{n-1})$$
$$q(z) = \langle u, y \rangle$$

Commitment Phase: For every coefficient u_i , the prover sends a Pedersen commitment

$$c_i = g^{r_i} h^{u_i}$$

Evaluation Phase: The verifier combines the coefficient commitments using the public vector $y(z)$

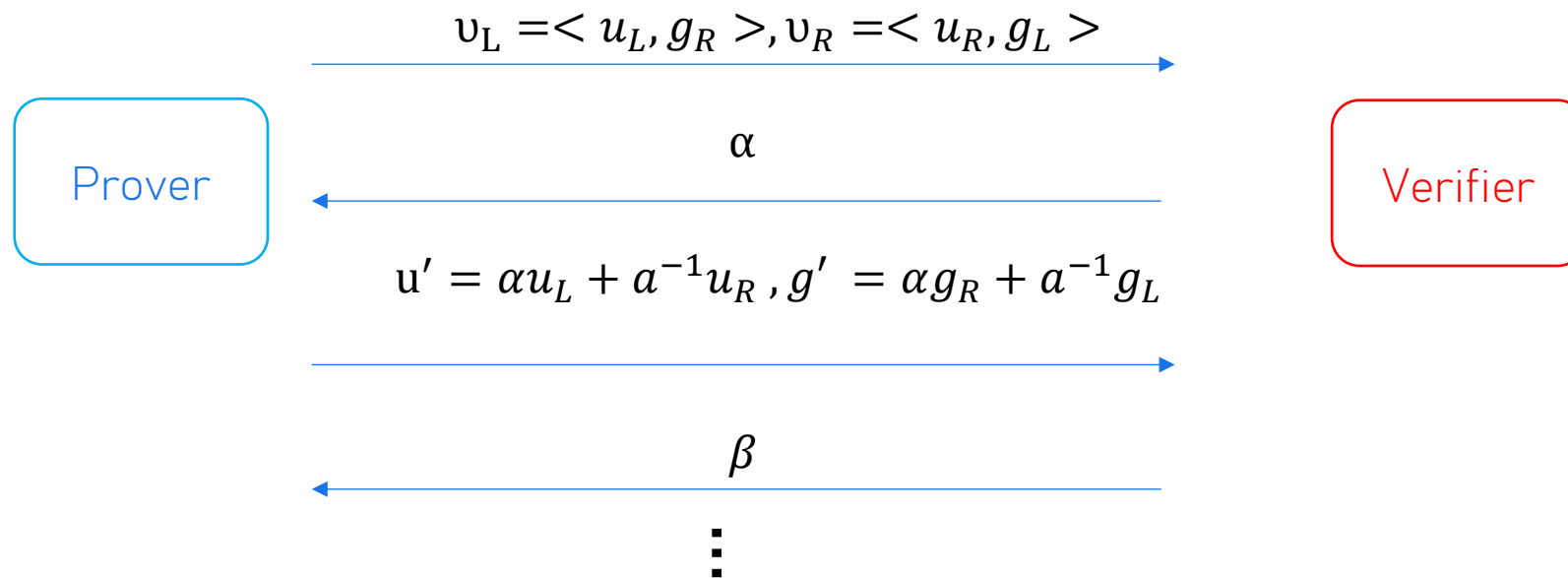
$$C = \prod_i c_i^{y_i} = \text{Commit} \left(\sum_i u_i y_i \right) = \text{Commit}(q(z))$$

Commitment size = $O(n)$

Bulletproofs

- Reduces the evaluation proof size of the polynomial commitments
- Uses a **vector commitment**: $\mathcal{C} = \langle u, g \rangle$ and use an **inner product argument**.

$$u = (u_L, u_R), \quad g = (g_L, g_R)$$



Commitment: $O(1)$, Evaluation Proof: $O(\log n)$